Exploratory Analysis and Assessment of Spatial Separability in Dark Matter Substructure Dataset

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Abstract. This work presents an exploratory analysis of a synthetic dataset of strong gravitational lensing images with three classes: no substructure, spherical substructure, and vortex substructure. While the dataset exhibits ideal structural properties (perfect class balance and uniform spatial resolution), our analysis reveals that class discrimination is not achievable through low-level statistical descriptors or linear projections. Both principal component analysis and clustering validity metrics indicate a near-complete overlap of class distributions in feature space. These findings suggest that the underlying discriminative structure is highly non-linear and spatially localized, motivating the use of deep convolutional architectures capable of hierarchical feature extraction.

1 Introduction

The precise nature of dark matter remains one of the most prominent open questions in modern astrophysics. Strong galaxy-galaxy lensing offers a unique observational window, as dark matter substructures within the primary lens halo perturb the lensed images. While deep learning models, particularly Convolutional Neural Networks (CNNs), have emerged as natural candidates for extracting these subtle perturbations , a rigorous assessment of the input data manifold is a crucial prerequisite. This work conducts a comprehensive exploratory data analysis (EDA) on a synthetic strong lensing dataset to evaluate spatial separability. We demonstrate that while the dataset exhibits ideal structural consistency, class discrimination is unattainable via low-level statistical features or linear projections, thereby formally justifying the need for deep hierarchical feature extraction.

2 Methodology

2.1 Dataset description

The dataset was synthetically generated utilizing the `lenstronomy` package. It comprises strong gravitational lensing images categorized into three distinct morphological classes: no substructure, subhalo (spherical) substructure, and vortex substructure.

2.2 Class Representativeness

Class imbalance is a well-known source of bias in supervised learning. To assess this risk, we compute the relative frequency (RF) of each class in both training and validation splits. The results show a perfectly uniform distribution (RF ≈ 0.33) ensuring that no class-dependent bias

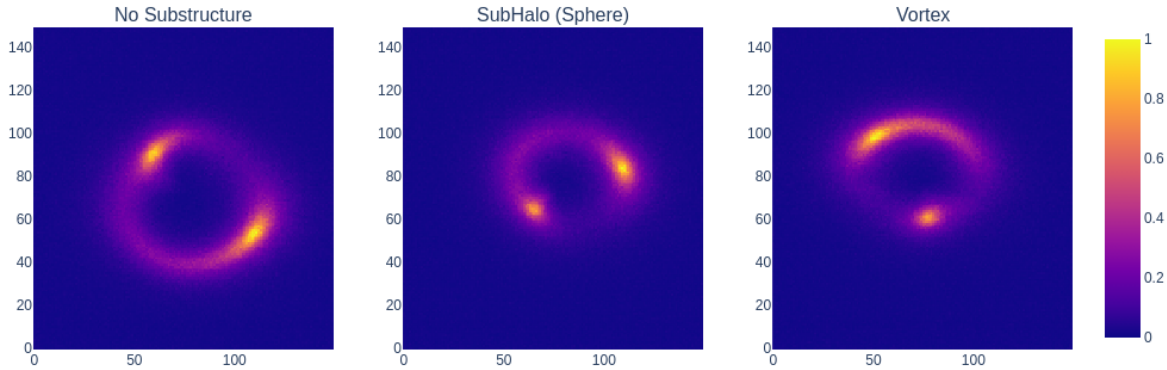

Figure 1: Example of a No Substructure, a Sphere, and a Vortex image.

is introduced during training. Consequently, no rebalancing strategies (e.g., class weighting or synthetic sampling) are required.

2.3 Spatial Consistency and Resolution

Variability in input geometric dimensions can introduce severe architectural constraints and noise in convolutional models. An exhaustive dimensional analysis confirms absolute structural homogeneity across the dataset: all samples invariably exhibit a spatial resolution of 150×150 pixels with a $1:1$ aspect ratio and a single grayscale channel. This structural invariance guarantees that no aggressive preprocessing - such as cropping, padding, or affine transformations - is required, thereby preserving the original spatial morphology of the lensing phenomena.

3 Spatial Separability Analysis

3.1 Low-Level Statistical Overlap

First-order statistical descriptors (mean μ and standard deviation σ) were extracted to evaluate the discriminative capacity of global intensity patterns. As illustrated in Figure 3, the feature space exhibits a severe, near-perfect overlap across all three classes. This empirical evidence demonstrates that global luminance and contrast variations are entirely decoupled from the presence of dark matter substructures.

3.2 Linear Subspace Analysis and Clustering Metrics

To formally quantify the high-dimensional entanglement, we projected the data onto a linear subspace utilizing the top 25 Principal Components. The clustering validity metrics strongly corroborate the initial low-level findings: The Silhouette Score yielded a value of -0.0022. This negative, near-zero magnitude mathematically confirms that samples frequently lie on or cross the decision boundaries of neighboring classes in the linear manifold. The Davies-Bouldin Index reached an exceptionally high value of 109.92, indicating an intra-class variance that overwhelmingly eclipses

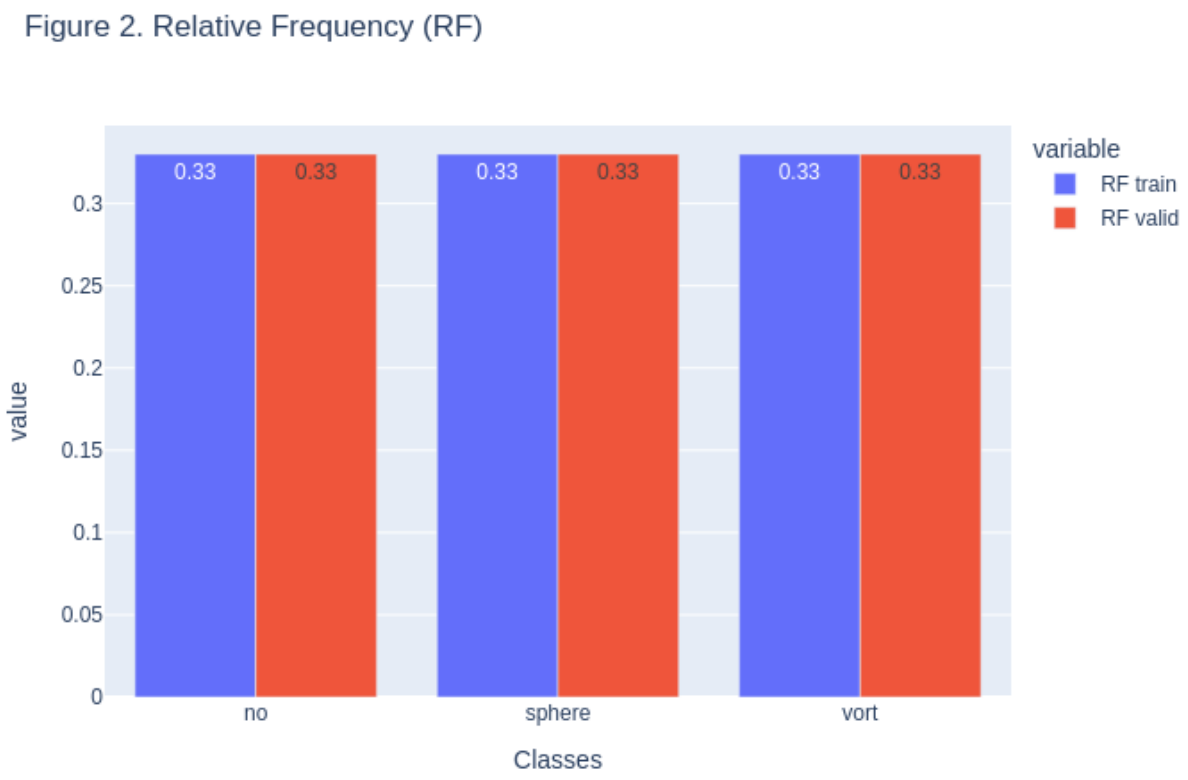

Figure 2: Relative frequency of each class.

inter-class dispersion. Finally, the Fisher Discriminant Ratio (FDR) across the principal components peaked at merely $\approx 800\mu$ (Figure 4). Even at its maximum, these vanishingly small FDR values definitively prove that the variance induced by identical macroscopic lensing topologies completely masks the localized signal of the subhalos in any linear projection.

4 Conclusion

In conclusion, while the dark matter substructure dataset provides a highly controlled and balanced environment (eliminating the need for morphological preprocessing or synthetic resampling), the intrinsic signal of the substructures (spheres and vortices) is deeply embedded within the global image variance. The failure of linear separability metrics (Silhouette ≈ 0, DB Index > 100, and vanishing Fisher Ratios) strictly rules out shallow learning techniques. Consequently, this EDA formally establishes the baseline complexity of the dataset. By demonstrating the inadequacy of shallow learning techniques , it provides the rigorous mathematical justification required for the future deployment of spatially-aware deep learning architectures, such as Convolutional Neural Networks, to isolate the localized perturbations induced by dark matter subhalos.

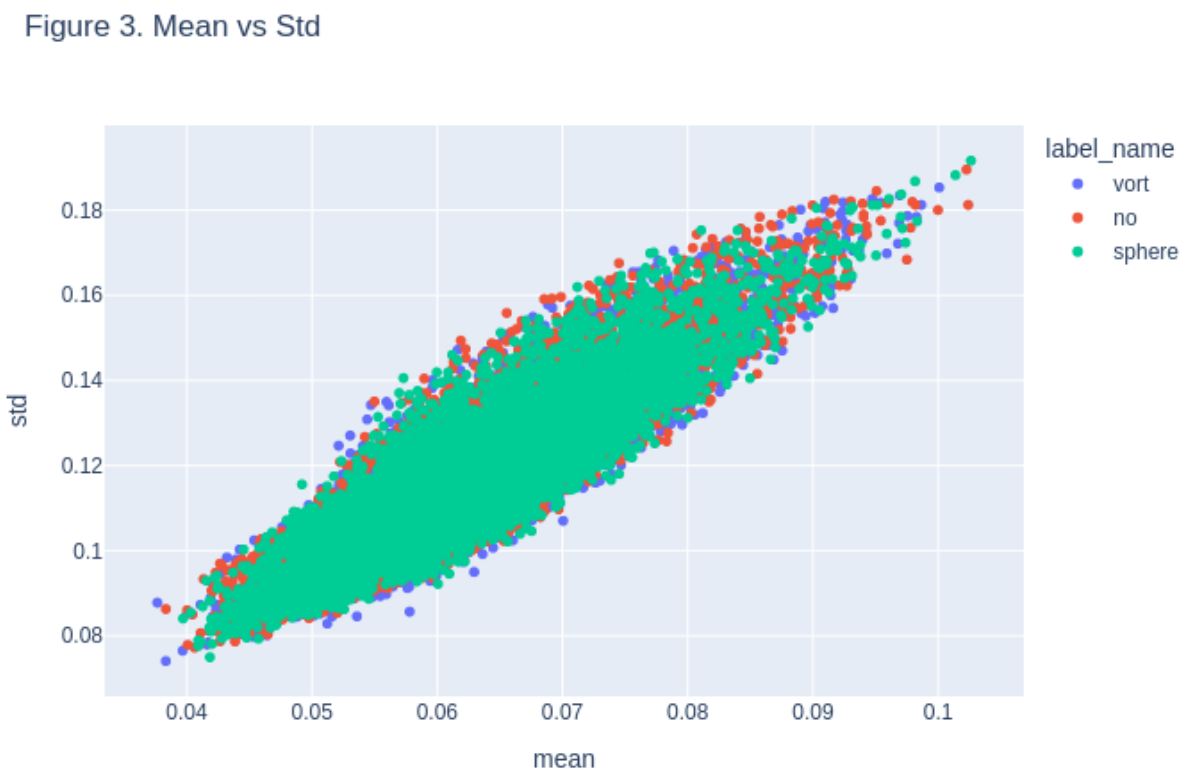

Figure 3: Mean and Standard Desviation of each class.

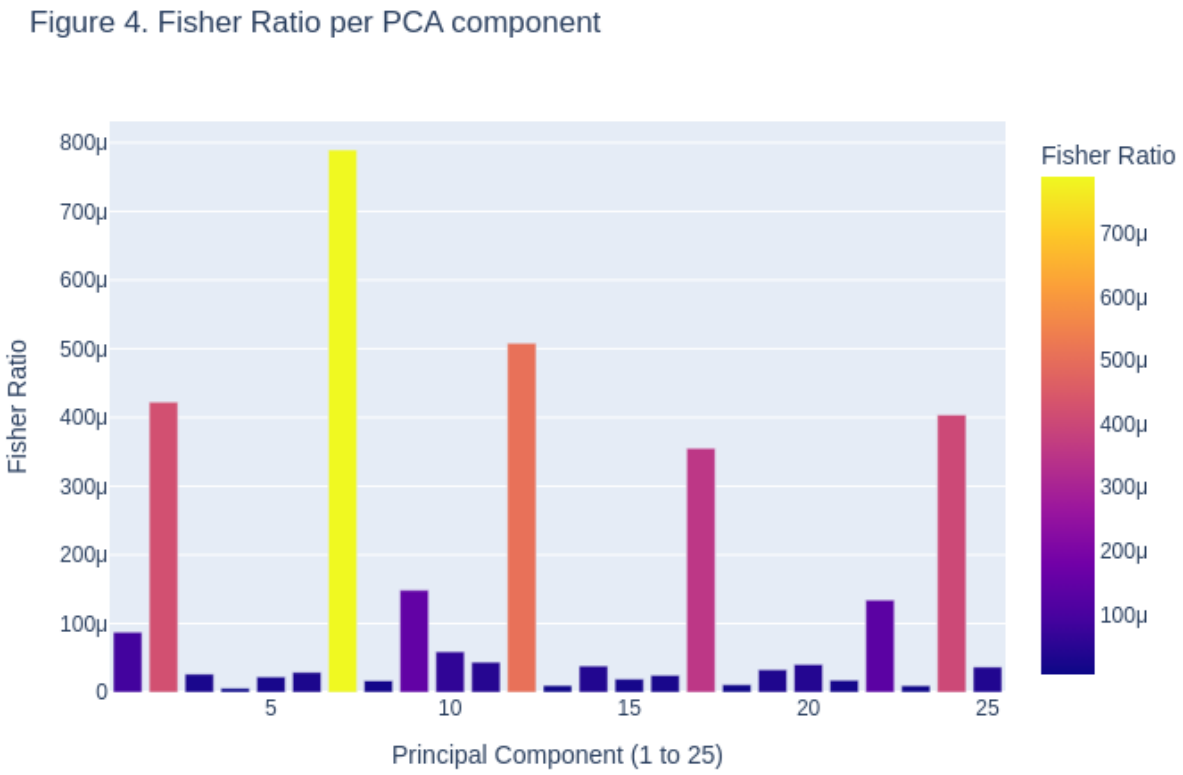

Figure 4: Fisher Ratio per PCA component.